

Measured Layer Sensitivity Guides Mixed-Precision ADC Allocation for CIM-Based LLM Inference

Anonymous Author(s)

Abstract

Compute-in-Memory (CIM) architectures accelerate large language model (LLM) inference by performing transformer-layer matrix-vector multiplications directly in resistive memory arrays, but their analog-to-digital converters (ADCs) dominate chip area. Mixed-precision ADC allocation can reduce this cost, yet conventional proxy signals—saturation rate and Hessian trace—do not reliably predict which transformer layers tolerate lower ADC precision in the CIM setting. We directly profile per-layer-type sensitivity under ADC bit reduction and uncover a *saturation-sensitivity inversion*: the highest-saturation attention layers (W_{qkv}) are the *least* accuracy-sensitive, while low-saturation FFN layers (W_{fc2}) are the most—the opposite of what conventional proxies suggest. Based on this finding, we propose a low-cost profiling-guided allocation flow that extracts the sensitivity signal in $G+1$ evaluation runs and feeds it to an area-aware integer linear program (ILP). On OPT-125M, the flow outperforms saturation-guided, Hessian-guided, and random baselines under the same ILP solver, achieving 20% ADC-area savings at under 1% relative perplexity degradation. On OPT-1.3B, it outperforms saturation-guided allocation at both tested budgets. Transfer sanity checks confirm that the sensitivity ordering generalizes to higher-bit deployment regimes with negligible accuracy cost.

CCS Concepts

• **Hardware** → **Emerging architectures**; **Electronic design automation**; *Emerging tools and methodologies*.

Keywords

Compute-in-memory, large language model, analog-to-digital converter, mixed-precision allocation, sensitivity profiling

1 Introduction

CIM architectures [5, 16] accelerate LLM inference by performing matrix-vector multiplications directly in resistive memory arrays. The ADCs that digitize the accumulated column currents grow exponentially in area with bit resolution: NeuroSIM [3] reports that ADCs occupy 24.4% of total chip area at 7-bit and 63.9% at 10-bit for OPT-125M [22]. Mixed-precision ADC allocation—assigning fewer bits to tolerant layers—can substantially reduce this cost. The central challenge is identifying which layers can tolerate reduced precision.

Existing digital mixed-precision methods [7, 18] rely on Hessian traces or saturation rates to guide bit allocation. In the CIM ADC regimes we study, however, these proxies fail to predict measured sensitivity: saturation rate is anti-correlated and Hessian trace is uncorrelated with directly measured

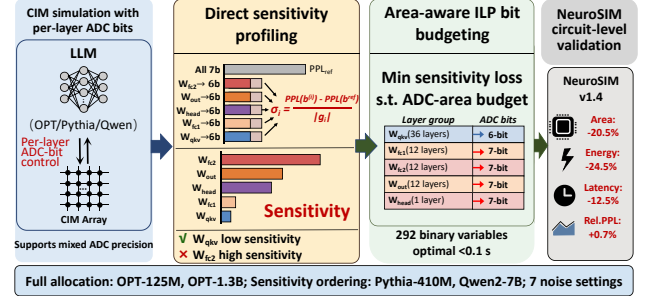


Figure 1: Proposed allocation flow. (1) CIM simulation evaluates LLM layers with configurable per-layer ADC precision. (2) Direct group-level profiling measures ΔPPL per layer group by lowering one group at a time. (3) An area-aware ILP assigns ADC bits under an ADC-area budget, reducing low-sensitivity W_{qkv} layers to 6-bit while keeping sensitive FFN/output groups at 7-bit. (4) NeuroSIM validates the resulting accuracy, ADC-area, energy, and latency trade-offs.

accuracy impact (Sec. 5). A conventional saturation-guided strategy would therefore protect the wrong layers.

We address this gap by directly measuring per-layer-type sensitivity under ADC bit reduction, revealing a *saturation-sensitivity inversion*: the highest-saturation layers (W_{qkv}) are the least sensitive, while the lowest-saturation layers (W_{fc2}) are the most. This measured signal is low-cost ($G+1$ evaluation runs) and transferable—low-bit profiling produces the same group-level ILP allocations when deployed at higher-bit regimes in all evaluated settings. Using this signal, the proposed allocation flow outperforms all tested proxy baselines under the same area budget (Fig. 1).

Contributions:

- (1) We show that saturation and Hessian proxies fail to predict ADC sensitivity in the CIM setting, and that direct profiling reveals an opposite ranking—validated on OPT-125M/1.3B and confirmed directionally on Pythia-410M and Qwen2-7B.
- (2) We propose a group-level profiling protocol (Algorithm 1) that extracts the allocation signal in $G+1$ evaluation runs. The resulting ordering is stable across four operating points and transfers to higher-bit deployment with zero group-level regret.
- (3) Under the same ILP solver, measured sensitivity consistently yields better allocations than all tested proxy signals on OPT-125M (three baselines) and OPT-1.3B (saturation baseline), including at practical higher-bit regimes.

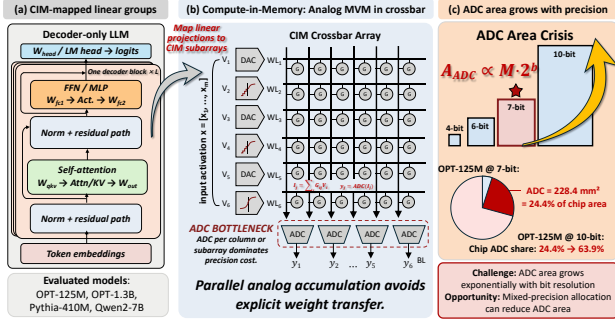


Figure 2: LLM-to-CIM mapping and ADC-area motivation. (a) A representative decoder-only LLM block is decomposed into CIM-mapped linear groups, including W_{qkv} , W_{out} , W_{fc1} , W_{fc2} , and W_{head} , which are the sensitivity groups profiled in this work. Split and fused-QKV implementations are represented by W_{qkv} , and gated-MLP projections are mapped to the corresponding FFN groups. (b) Linear projections are mapped onto CIM subarrays, where analog column-current accumulation performs MVM and ADCs digitize the outputs. (c) ADC area scales with the number of columns and bit resolution ($A_{ADC} \propto M \cdot 2^b$), making uniform high-precision ADCs costly and motivating mixed-precision ADC allocation.

2 Background and Related Work

2.1 CIM Array Architecture and ADC Bottleneck

Fig. 2a groups decoder-only LLM linear projections into the sensitivity groups used in this work, while Fig. 2b shows their mapping to CIM subarrays. In a CIM crossbar, weights W are programmed as conductance states in resistive memory cells. The input vector x is applied row-wise through DACs, and column currents accumulate according to Kirchhoff’s current law before being digitized by ADCs. As illustrated in Fig. 2c, for an M -column subarray with a b -bit MLSA-type ADC [3], the ADC area scales as

$$A_{ADC} \propto M \cdot 2^b \quad (1)$$

because the multi-level sense amplifier requires 2^b comparator levels. Across all 73 linear layers of OPT-125M (4 unique weight-matrix shapes), this exponential scaling produces the area breakdown shown in Fig. 3, where the ADC share grows from 24% at 7-bit to 64% at 10-bit.

2.2 Activation Outliers and ADC Saturation

LLM activations exhibit systematic *outlier channels* [6]: certain feature dimensions consistently carry magnitudes far exceeding the median. As shown in Fig. 4, saturation rates vary dramatically across layer types— W_{qkv}/W_{fc1} saturate

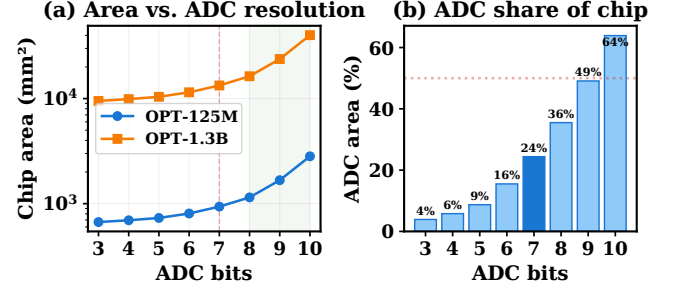


Figure 3: (a) Chip area vs. ADC resolution (NeuroSIM): 2^b scaling drives a $3\times$ increase from 7b to 10b. (b) ADC share of total chip area: ADC dominates ($>50\%$) above 9-bit.

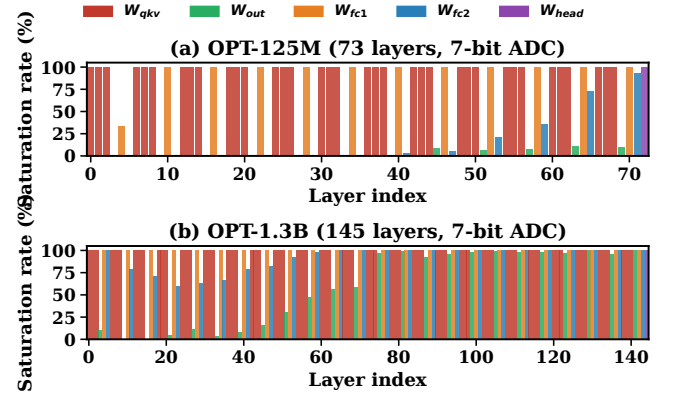


Figure 4: Per-layer max-clip ADC saturation rate at 7-bit. (a) OPT-125M (73 layers): W_{qkv}/W_{fc1} saturation rate near 100%. (b) OPT-1.3B (145 layers): the same pattern persists. In both models, W_{out}/W_{fc2} remain below 20%.

near 100% while W_{out}/W_{fc2} remain below 20%—and this pattern persists from OPT-125M to OPT-1.3B. A widely used proxy for layer difficulty is the *max-clip saturation rate*:

$$s_i^{\max} = \Pr[y_{ij} > V_{FS}^{\max}] \quad (2)$$

This proxy (Eq. 2) suggests that layers with high $s_i^{\max}$ require more ADC bits. SmoothQuant [20] mitigates outliers by migrating activation difficulty to weights. However, as we demonstrate in Sec. 5, $s_i^{\max}$ does not reliably predict the accuracy impact of reducing ADC resolution.

2.3 Related Work

CIM for LLMs. PRIME [5] and ISAAC [16] pioneered CIM for CNNs; recent surveys [19] identify ADC as the bottleneck when scaling CIM to LLMs. FlexCIM [15] and P3-LLM [4] address LLM CIM inference but do not optimize per-layer ADC precision.

ADC-Aware CIM. Azamat et al. [1] and CIM²PQ [17] apply mixed-precision ADC to CNN accelerators. Chen et

al. [2] employ Hessian-based sensitivity for ReRAM CIM; in our LLM ADC setting, Hessian trace is uncorrelated ($\rho=0.20$) and saturation is anti-correlated ($\rho=-0.70$) with measured sensitivity. Khwa et al. [10] and MXFormer [9] validate mixed-precision CIM at processor level.

LLM Quantization. SmoothQuant [20], GPTQ [8], and AWQ [12] target digital accelerators. HAWQ [7] and HAQ [18] use Hessian traces or RL for digital quantization; these proxies do not transfer to CIM ADC noise. Yan et al. [21] address worst-case NVCiM accuracy through training-time noise injection; our work complements theirs with deployment-time allocation. We employ DNN+NeuroSIM [3, 11, 14] for circuit-level validation.

3 Sensitivity Profiling Methodology

3.1 Measurement Protocol

We define the *measured sensitivity* of layer group i as the perplexity increase caused by reducing its ADC resolution:

$$\sigma_i = \frac{\text{PPL}(\mathbf{b}^{(i)}) - \text{PPL}(\mathbf{b}^{\text{ref}})}{|g_i|} \quad (3)$$

where $\mathbf{b}^{\text{ref}} = [b_{\text{nom}}, \dots, b_{\text{nom}}]$ is the uniform baseline and $\mathbf{b}^{(i)}$ reduces only group i to b_{prb} while keeping all other layers at b_{nom} . Fig. 5 illustrates our qualitative hypothesis that different linear groups induce distinct noise-propagation paths through a decoder block. For efficiency, we adopt group-level measurement (Eq. 3)—reducing all layers of the same functional type simultaneously—yielding only $G = 5$ measurements rather than 73. Algorithm 1 formalizes the protocol.

3.2 Group-Level Granularity Validation

We validate group-level profiling with depth-bin analysis and per-layer spot-checks on three representative blocks (Fig. 6). FFN layers are consistently more sensitive than attention layers at every depth position; sensitivity generally decreases with depth, though W_{fc2} remains above 0.4 ΔPPL even at Late positions. We further compare group-ILP against per-layer ILP (73 individual sensitivity measurements): group-ILP outperforms per-layer ILP at moderate budgets ($\leq 30\%$ savings) because group-level averaging reduces measurement noise from limited evaluation batches. Per-layer ILP recovers its advantage only at aggressive budgets ($>40\%$).

We partition the WikiText-2 [13] evaluation set into three non-overlapping splits: $\mathcal{D}_{\text{cal}}$ (4 batches) for ADC full-scale calibration, $\mathcal{D}_{\text{prof}}$ (10 batches, $\approx 5\text{K}$ tokens) for sensitivity profiling (Algorithm 1), and $\mathcal{D}_{\text{test}}$ (100 batches, $\approx 51\text{K}$ tokens) for all final PPL evaluations reported in Table 3. All sequences are 512 tokens with p99-clip ADC calibration. The profiling and test sets do not overlap; allocation decisions are made solely on $\mathcal{D}_{\text{prof}}$.

3.3 Robustness of Measured Sensitivity

Table 1 summarizes robustness along five axes. The ILP assignment is invariant across calibration, operating-point, and transfer conditions in this group-level search space; under enhanced noise models, 5 of 7 configurations preserve the

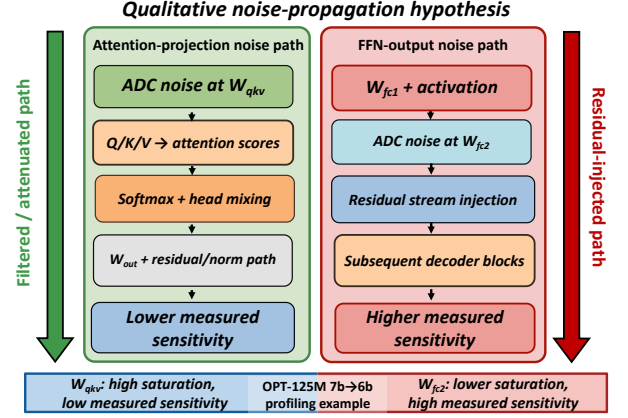


Figure 5: Qualitative noise-propagation hypothesis for the saturation-sensitivity inversion. Noise injected at W_{qkv} passes through attention-score computation, softmax/head mixing, W_{out} , and residual/normalization paths before affecting later states, which is consistent with lower measured sensitivity in the OPT-125M 7b $\rightarrow$ 6b profiling example. In contrast, noise injected at W_{fc2} enters the residual stream after the FFN and can affect subsequent decoder blocks, which is consistent with higher measured sensitivity. The diagram illustrates an interpretation of the measured trend rather than a formal proof.

Algorithm 1 Group-Level ADC Sensitivity Measurement

Require: Model $\mathcal{M}$, calibration data $\mathcal{D}_{\text{cal}}$, profiling data $\mathcal{D}_{\text{prof}}$, nominal bits b_{nom} , probe bits b_{prb}
Ensure: Per-group measured sensitivity $\{\sigma_g\}_{g=1}^G$
1: Partition N layers into G groups by functional type
2: Calibrate ADC full-scale $V_{\text{FS},i}$ for each layer i using $\mathcal{D}_{\text{cal}}$
3: $\mathbf{b}^{\text{ref}} \leftarrow [b_{\text{nom}}] \times N$
4: $\text{PPL}_{\text{ref}} \leftarrow \text{Eval}(\mathcal{M}, \mathbf{b}^{\text{ref}}, \mathcal{D}_{\text{prof}})$
5: **for** each group $g = 1, \dots, G$ **do**
6: $\mathbf{b}^{(g)} \leftarrow \mathbf{b}^{\text{ref}}$; set $b_i^{(g)} = b_{\text{prb}}$ for all $i \in g$
7: $\text{PPL}_g \leftarrow \text{Eval}(\mathcal{M}, \mathbf{b}^{(g)}, \mathcal{D}_{\text{prof}})$
8: $\sigma_g \leftarrow (\text{PPL}_g - \text{PPL}_{\text{ref}})/|g|$
9: **end for**
10: **return** $\{\sigma_g\}$

same assignment (Table 1). W_{qkv} remains a low-sensitivity group in the evaluated OPT settings.

4 Area-Aware Bit Budgeting

We formulate the budgeted bit assignment as an integer linear program (ILP) that integrates measured sensitivities, layer-specific area coefficients, and bit-choice constraints into a single optimization step.

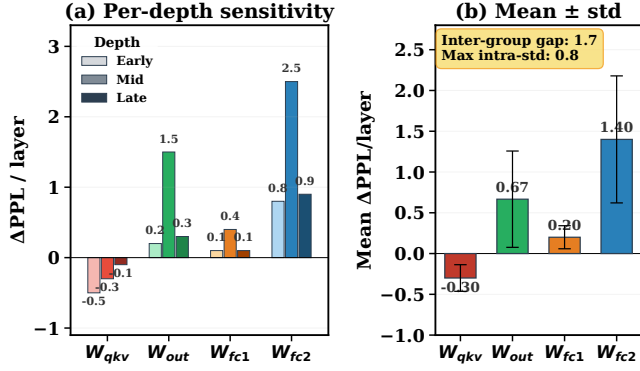


Figure 6: Group-level profiling validity. (a) Per-depth sensitivity for each layer type: all W_{qkv} bins are negative while all W_{fc2} bins are positive, confirming the inter-group ordering at every depth position. (b) Mean $\pm$ std across depth bins.

Table 1: Robustness summary. “Assignment stable?” = ILP produces the same protected-group set across all tested conditions.

Axis	Conditions	Stable?
Calib. seeds	5 offsets	Yes (5/5)
Calib. scheme	p99 / p99.5 / max	Yes (3/3)
Operating pt.	7→6 / 8→7 / 9→8 / 10→9	Yes (4/4)
Transfer	Profile@7, deploy@8/9/10	Yes (3/3)
Noise model	ADC + 6 thermal/device variations	Partial (5/7)

4.1 ILP Formulation

Each layer inherits the measured sensitivity of its functional group: $\sigma_i = \sigma_g$ for all layers $i \in g$. Given N layers with these sensitivities and a target ADC area budget B , we minimize total sensitivity-weighted degradation:

$$\min_{\mathbf{b}} \sum_i \max(0, b_{\text{nom}} - b_i) \cdot \sigma_i^+ \quad \text{s.t.} \quad \sum_i a_i 2^{b_i} \leq B, \quad b_i \in \mathcal{B} \quad (4)$$

where $\sigma_i^+ = \max(\sigma_i, 0)$ clamps negative sensitivities to zero (negative values can arise from evaluation noise and would otherwise cause the solver to exploit finite-sample variance), b_{nom} is the nominal bit width (7 for profiling, 10 for deployment), $\mathcal{B}$ is the set of allowed bit widths ($\{4, 5, 6, 7\}$ for the 7b stress-test regime; $\{7, 8, 9, 10\}$ for 10b deployment; in practice the ILP selects only 6 and 7 at the tested moderate budgets), a_i is the layer-specific ADC area coefficient proportional to the number of ADC columns in the NeuroSIM mapping, and the area constraint follows the MLSA model (Eq. 1). The resulting mixed-precision ADC-area saving is

$$\Delta A_{\text{ADC}} = \sum_i a_i (2^{b_{\text{nom}}} - 2^{b_i^*}) \quad (5)$$

We linearize (4) via binary indicator variables $x_{i,b} \in \{0, 1\}$ and solve with `scipy.optimize.milp`. For OPT-125M (292 binary variables), the ILP solves to proven optimality in

Table 2: Surrogate validation. Top: allocation optimality (ILP matches oracle). Bottom: representative pairwise interaction tests.

<i>Allocation optimality (20.5% savings)</i>		
ILP allocation PPL	308.6	(+0.7%)
Brute-force oracle PPL	308.6	(+0.7%)
Allocation regret	0.0 PPL	
<i>Pairwise interaction (representative pairs)</i>		
	Pred. Δ	Meas. Δ
$W_{out}+W_{fc2}$	+11.4	+11.7
$W_{qkv}+W_{fc2}$	+10.9	+9.4
$W_{fc1}+W_{head}$	+9.4	+10.7

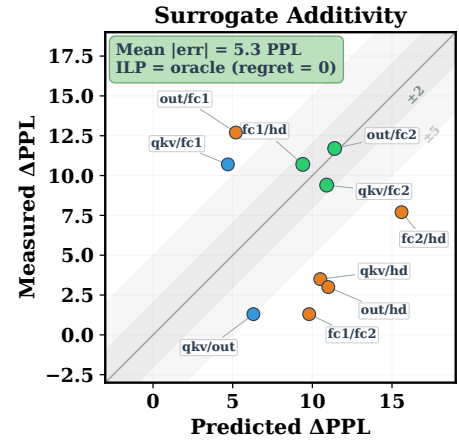


Figure 7: Surrogate additivity: predicted vs. measured pairwise Δ PPL. Green region = overestimate (conservative); red = underestimate. Mean |error| = 5.3 PPL, yet the ILP allocation still matches the brute-force oracle (regret = 0.0).

under 0.1 s. Exhaustive enumeration of all $4^5=1,024$ group-level configurations confirms that the ILP solution matches the brute-force optimum.

4.2 Surrogate Validation

The linear surrogate is approximate for pairwise Δ PPL prediction (Table 2, Fig. 7), but its primary role is *allocation ranking*. Two strong validation signals confirm its correctness: (1) exhaustive enumeration verifies that the ILP matches the brute-force optimum (regret = 0.0); (2) end-to-end measured PPL independently validates allocation quality. The surrogate suffices because moderate-budget ILP reduces only one or two groups, where individual measurements are exact by construction.

4.3 Computational Cost

Sensitivity measurement requires $G+1 = 6$ evaluation runs; each run evaluates the full $\mathcal{D}_{\text{prof}}$ split of 10 batches ($\approx 5K$

tokens). The ILP involves $N \times |\mathcal{B}|$ binary variables (292 for OPT-125M, <0.1 s; 580 for OPT-1.3B, <0.2 s). The total profiling overhead is negligible compared to deployment.

5 Experimental Evaluation

We validate the complete allocation pipeline with NeuroSIM [3] circuit-level analysis (45 nm CMOS, 128×128 RRAM cross-bar, MLSA-type ADC, 1-bit DAC input, INT8 weight mapping with bit-slicing), covering area, energy, latency, and downstream task accuracy.

5.1 Setup and Reference Definitions

We define four reference operating points. **FP32**: no CIM noise or weight quantization. **CIM-7b**: INT8 weights + 7-bit ADC with p99-clip calibration. **Profiling regime**: CIM-7b $\rightarrow$ 6b, where sensitivity contrasts are largest. **Deployment regime**: CIM-8b/9b/10b, where practical chips operate. The FP32-to-CIM gap is dominated by INT8 weight quantization without fine-tuning, not ADC noise. To confirm this, we measure an INT8-only control (12-bit ADC, effectively removing ADC noise) alongside CIM-7b and CIM-10b on the same 10-batch split used for the cross-regime sanity check in Fig. 8; absolute PPL values differ from the 100-batch $\mathcal{D}_{\text{test}}$ of Table 3 due to the different evaluation split, but the *relative* INT8-vs-ADC comparison is valid within this matched protocol:

Configuration	PPL
FP32 (no quantization)	36.2
INT8-only (12b ADC)	316.9
INT8 + 10b ADC	316.9
INT8 + 7b ADC	317.1

The INT8-only and CIM-7b baselines differ by only 0.2 PPL, indicating that ADC clipping contributes negligibly compared with the INT8 weight-quantization gap. Our goal is not to recover the INT8 quantization loss, but to reduce ADC cost under a fixed CIM quantization floor; improved weight quantization (e.g., GPTQ, AWQ) is orthogonal and can be combined with our allocation flow. CIM-7b through CIM-10b baselines are nearly identical under a consistent evaluation protocol (Fig. 8a); this 10-batch cross-regime sweep is used only to validate baseline stability, while all final allocation comparisons are reported on the 100-batch $\mathcal{D}_{\text{test}}$ in Table 3. The allocation problem we address—reducing ADC area with minimal *additional* degradation beyond the weight quantization floor—remains practically relevant because the relative PPL differences between allocation methods are consistent and reproducible, and the area savings scale to hundreds of mm^2 at practical model sizes.

5.2 Sensitivity Inversion and Cross-Regime Transfer

The measured sensitivities (Fig. 9) reveal that saturation rate ranks W_{qkv} and W_{fc1} as the most problematic layers ($\approx 100\%$ saturation), yet measured sensitivity ranks them *last* (0.13 and 0.20 $\Delta\text{PPL}/\text{layer}$ at 7b $\rightarrow$ 6b). W_{fc2} ranks first at 1.41

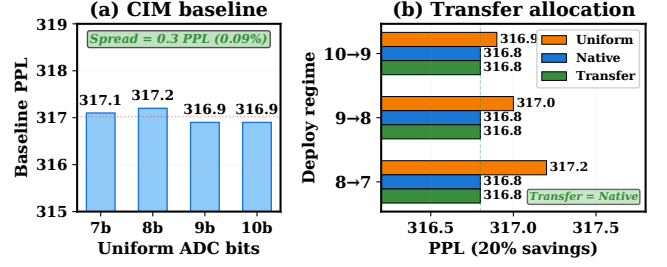


Figure 8: (a) CIM baseline PPL is nearly identical across 7–10b ADC (spread = 0.3 PPL; weight quantization dominates); evaluated on a consistent 10-batch protocol across all four regimes. (b) Transfer allocation: 7b $\rightarrow$ 6b profiling produces the same group-level ILP assignments as native profiling at all three tested higher-bit regimes. This 10-batch sweep validates cross-regime baseline stability; all final allocation comparisons use the 100-batch $\mathcal{D}_{\text{test}}$ in Table 3.

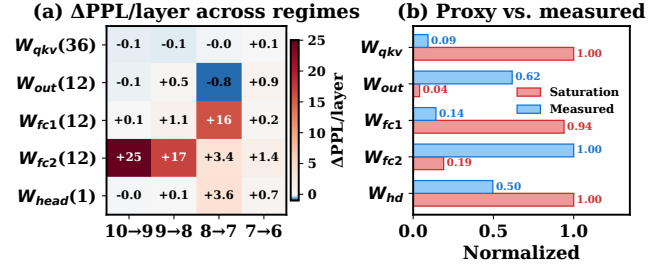


Figure 9: (a) Measured sensitivity ($\Delta\text{PPL}/\text{layer}$) across four operating points (OPT-125M). W_{fc2} is the most sensitive at 3/4 regimes; W_{qkv} is consistently the least. (b) Saturation proxy vs. measured rank: the proxy would protect the *wrong* layers ($\rho = -0.70$).

$\Delta\text{PPL}/\text{layer}$ despite only 19% saturation—an $11 \times$ inversion. The exact rank correlation varies across adjacent regimes; for non-adjacent and high-bit regime pairs, Spearman ρ ranges from 0.60 to 0.90. More importantly, the allocation-critical ordering is stable: W_{qkv} remains in the low-sensitivity set and FFN layers remain more sensitive than attention layers at every tested operating point.

We attribute this inversion to qualitative noise-propagation effects rather than treating them as formal proofs (Fig. 5). For W_{qkv} , ADC perturbations pass through attention-score computation, softmax/head mixing, W_{out} , and residual/normalization paths before affecting later hidden states, which can attenuate their measured impact. For W_{fc2} , perturbations are injected into the residual stream after the FFN and can influence subsequent decoder blocks, consistent with the higher measured sensitivity and the mid-depth sensitivity peak in Fig. 6a.

The FFN $>$ attention hierarchy holds at all four tested operating points (Fig. 9a). Fig. 10 visualizes the rank stability

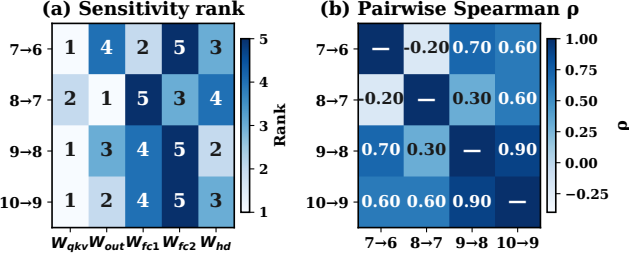


Figure 10: (a) Sensitivity rank across four operating points: W_{qkv} remains in the low-sensitivity set and the FFN > attention ordering is preserved throughout. (b) Pairwise Spearman ρ between regimes (0.60–0.90 for non-adjacent pairs).

and assignment overlap: 7b→6b profiling produces 5/5 group-level agreement with native profiling at all three tested higher-bit regimes, with zero group-level allocation regret (PPL within 0.2% of baseline).

5.3 Allocation Results

Table 3 reports measured PPL for all configurations. To isolate the value of the measured sensitivity *signal* from the ILP *solver*, we run four allocation signals through the same ILP at identical budgets. At 20.5% ADC-area savings, profiling-guided ILP (+0.7%) outperforms Hessian-ILP (+1.1%), saturation-ILP (+0.9%), and random allocation (+2.6%±1.4%, 20 trials). For the proxy baselines, we estimate per-group Hessian trace via diagonal Fisher approximation on $\mathcal{D}_{\text{prof}}$ using the same 10 profiling batches, normalize across groups, and feed the resulting proxy scores to the identical ILP solver; saturation-ILP uses per-group saturation rate as the proxy signal. The same pattern persists at 30% savings. Exhaustive group-level brute-force search confirms that the ILP solution matches the globally optimal group-level configuration. Fig. 11 visualizes the per-layer assignments; Fig. 12 shows the PPL–area Pareto frontier.

In a separate sanity check at the 10-bit deployment regime, CIM-7b through CIM-10b baselines yield nearly identical PPL (spread < 0.2%), confirming that ADC precision contributes negligibly beyond the weight quantization floor. Profiling-guided ILP at 10b saves 20% ADC-area (361 mm² via Eq. 5) with negligible PPL change, because the ILP concentrates the ADC budget on sensitive layers while reducing insensitive W_{qkv} layers. Under the same NeuroSIM mapping, the 20.5% ADC-area-saving allocation reduces NeuroSIM-reported ADC-path energy by 24.5% and ADC-read latency by 12.5%. SmoothQuant [20] is an orthogonal activation pre-processing step; it reduces outlier magnitude before ADC conversion and can be combined with profiling-guided allocation by re-profiling sensitivity on the SQ-transformed model. We report SQ+6b as a standalone reference; the combination SQ + profiling-guided ILP is a natural extension left to future work. The proxy ranking analysis (Fig. 13) further

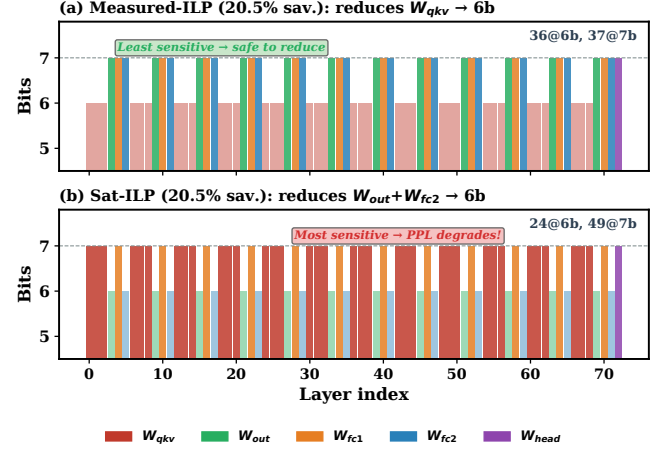


Figure 11: Per-layer bit assignment at 20.5% ADC savings. (a) Measured-ILP reduces W_{qkv} (least sensitive) to 6-bit. (b) Sat-ILP reduces W_{out}/W_{fc2} (most sensitive) to 6-bit—protecting the *wrong* layers. Same ILP solver, different signal, different allocation.

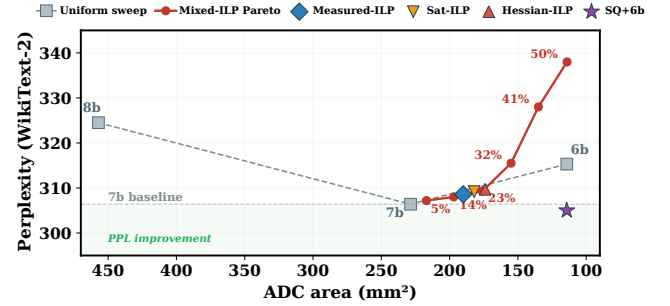


Figure 12: PPL vs. ADC area Pareto frontier. Mixed-ILP (red) achieves a better trade-off than the uniform-bit sweep (gray) at all evaluated budgets. SQ+6b is shown as an orthogonal activation-rescaling reference, not an ADC-allocation baseline.

confirms that saturation ($\rho = -0.70$) and Hessian ($\rho = 0.20$) are anti-correlated or uncorrelated with measured sensitivity.

Downstream task accuracy. We evaluate on five zero-shot tasks (570–3,000 samples each) with 95% bootstrap confidence intervals (Fig. 14). ILP-20% preserves CIM-7b accuracy within the confidence interval on all tasks ($|\Delta| = 0.2$ pp).

5.4 Cross-Model Validation and Noise Robustness

Table 4 reports sensitivity-ordering checks across four models and full allocation results on OPT-1.3B. OPT-1.3B allocation confirms the practical value at scale: profiling-guided ILP outperforms saturation-ILP at both 20% and 30% savings.

Table 3: Allocation results across multiple signals and budgets (OPT-125M, 7b regime). All rows use p99 calibration and 100-batch $\mathcal{D}_{\text{test}}$. Sav.% = ADC-area saving relative to uniform 7b; Rel. = relative PPL degradation vs. uniform 7b; Area = total chip area (mm^2). SQ+6b is an orthogonal activation-rescaling reference, not an ADC-allocation baseline. Random (20) reports mean $\pm$ std over 20 independent random allocations.

Signal	Solver	Sav.%	PPL	Rel.	Area
<i>Uniform baselines</i>					
—	7b	0	306.4	—	937
—	6b	50	315.3	+2.9%	822
<i>20.5% savings</i>					
Measured	ILP	20.5	308.6	+0.7%	890
Saturation	ILP	20.5	309.2	+0.9%	890
Hessian	ILP	20.5	309.7	+1.1%	890
Random (20)	—	20.5	314.5 $\pm$ 4.4	+2.6% $\pm$ 1.4%	890
Grp-Oracle	BF	20.5	308.6	+0.7%	890
<i>30% savings</i>					
Measured	ILP	30	309.3	+0.9%	861
Saturation	ILP	30	310.5	+1.3%	861
Hessian	ILP	30	309.5	+1.0%	861
<i>Orthogonal reference</i>					
—	SQ+6b	50	305.0	−0.5%	822

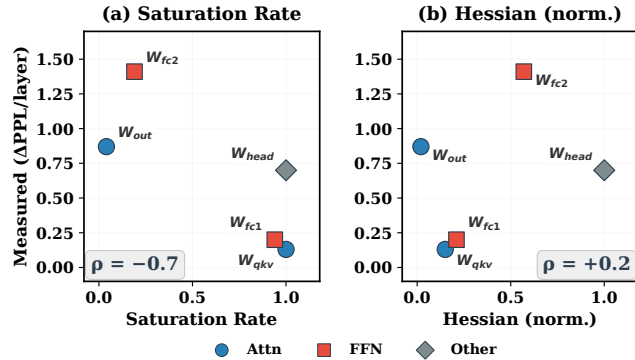


Figure 13: Proxy failure: saturation rate (a) and Hessian trace (b) vs. measured sensitivity. Both proxies would protect the *wrong* layers.

We further augment the ADC clipping model with additive thermal noise and multiplicative device variation (six configurations). W_{fc2} remains the most sensitive non- W_{head} layer in 5/7 settings (Fig. 15); the ILP assignment is stable in 5/7 configurations (Table 1).

6 Conclusion

We have shown that directly measured layer sensitivity provides a more reliable ADC allocation signal than saturation or Hessian proxies in the CIM regimes tested. The key empirical

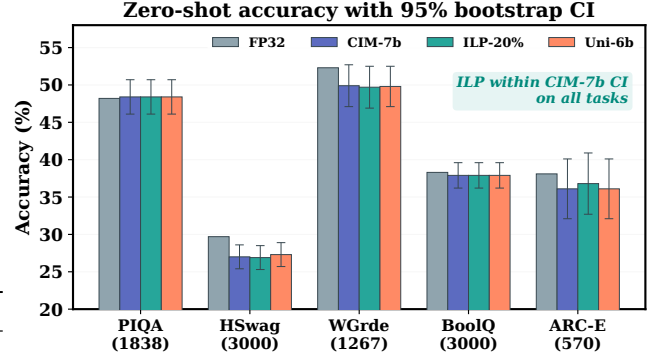


Figure 14: Zero-shot accuracy with 95% bootstrap CI. ILP-20% (green) falls within the CIM-7b confidence interval on every task. Mean $|\Delta| = 0.2$ pp.

Table 4: Cross-model validation. Top: sensitivity-ordering checks across four models (Pythia/Qwen report ordering only). Bottom: full allocation results on OPT-1.3B (145 layers).

<i>Sensitivity ordering</i>		Probe	Most	Least	FFN>Att?
Model	Arch.				
OPT-125M	OPT	7→6	W_{fc2}	W_{qkv}	Yes
OPT-1.3B	OPT	7→6	W_{fc2}	W_{qkv}	Yes
Pythia-410M	NeoX	7→6	FFN	Attn	Yes
Qwen2-7B	Qwen2	8→7	W_{fc1}	W_{out}	Yes
<i>OPT-1.3B allocation (7b baseline = 672.6)</i>					
Signal / Budget		PPL	Rel.		
Profiling / 20%		672.2	−0.06%		
Sat-ILP / 20%		675.5	+0.43%		
Profiling / 30%		674.7	+0.31%		
Sat-ILP / 30%		678.6	+0.89%		

finding—a saturation-sensitivity inversion where the highest-saturation layers are the *least* accuracy-sensitive—can be extracted in $G+1$ evaluation runs and remains stable across calibration seeds, operating points, and transfer regimes. On OPT-125M, profiling-guided ILP outperforms saturation-, Hessian-, and random-guided baselines under the same solver, achieving 20% ADC-area savings at +0.7% relative PPL (Table 3). On OPT-1.3B, it outperforms saturation-guided allocation at both tested budgets. Transfer sanity checks confirm that the sensitivity ordering generalizes to higher-bit deployment regimes with negligible accuracy cost. The practical payoff is ADC-area reduction that scales from tens of mm^2 at OPT-125M to hundreds at OPT-1.3B.

Limitations. The linear surrogate preserves correct allocation ranking at moderate (10–30%) savings but may degrade at aggressive budgets where multi-group interactions become significant. Group-level profiling is a first-order signal; finer depth-aware refinement may further improve allocation quality. Better weight quantization (e.g., GPTQ, AWQ) would reduce the absolute FP32-to-CIM gap but would not invalidate the ADC allocation comparison, which operates on relative sensitivity differences above the quantization floor.

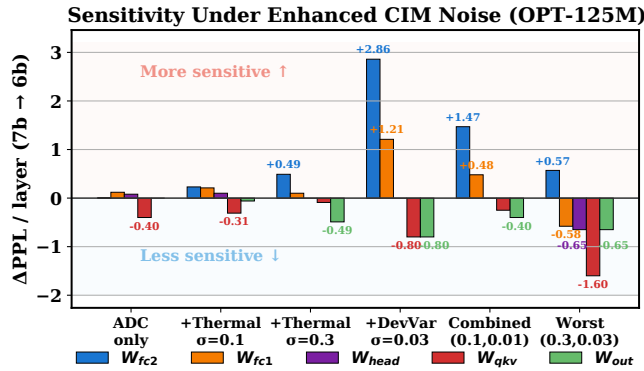


Figure 15: Sensitivity robustness under six enhanced CIM noise configurations (thermal + device variation). W_{fc2} remains the most sensitive non- W_{head} layer in 5/7 settings; under the strongest device variation ($\sigma_{dev} = 0.03$), W_{fc2} exhibits the largest positive ΔPPL (+2.86/layer).

All experiments use NeuroSIM simulation; silicon validation and extension to encoder-decoder or MoE architectures remain future work.

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